MESS PROVIDER

PRODUCT + TECHNICAL BLUEPRINT

Provider mobile app • Admin operations console • Shared backend

Product principle

Every provider can configure their own menu, portion quantities, plan model, meal slots, special dishes and extra charges. Shared backend rules keep all provider variants safe and consistent.

Working handoff for Antigravity implementation
25 October 2026 | Draft for engineering review

1. Product vision and scope

MESS is a subscription and in-person meal-redemption platform for student messes, restaurants, canteens and home-tiffin providers. The provider registers a real business location, defines service times and menus, configures plans, then displays a provider-generated daily QR at the counter. The customer scans that QR to mark a meal.

Current release boundary

Offline meal collection only. Delivery is out of scope. The customer scans the provider QR, never the other way around.

Provider types

Type	Typical use	Required setup
Student mess	Recurring lunch/dinner or multiple daily slots	Location, schedule, plan rules, base thali and extras
Hotel / restaurant	Dine-in subscription and daily meals	Opening hours, menu, plan slots, daily prices
Home tiffin service	Home-cooked food served from a listed point	Owner identity, verified address and actual serving arrangements
Hostel / college canteen	Campus food operation	Campus location, opening calendar, meal slots and service limits

Product goals

- Easy provider registration and approval workflow with manual location entry and optional map pin.
- Flexible plan configuration without custom development for every mess.
- Accurate menu, portion, special-food and extra-serving rules.
- Counter-friendly daily QR, customer eligibility checks and a complete redemption history.
- Provider reporting that distinguishes gross sales, fees, refunds, payable amount and actually settled money.
- Restricted admin console for provider review, support, finance, audit and developer operations.

Non-goals for first release

- Delivery routing, rider assignment and live delivery tracking.
- Fake ratings, fake customer counts or fabricated analytics.
- Local/client-side QR tokens that can redeem production meals.
- Provider-controlled approval status or client-side financial writes.

2. Provider registration and first setup

The onboarding wizard should feel short and guided. Keep advanced options grouped by step and allow a draft to be resumed. A provider is not discoverable and cannot accept live redemptions until required review and publication conditions pass.

Step	Screen / decisions	Server-side outcome
1	Create owner account: email/phone + password or Google OAuth	Authenticated profile; verify contact; protect sessions
2	Business type, name, owner contact, phone/email, applicable food registration	Draft business with owner association; required fields validated
3	Address, locality, city, postal code, optional map pin, opening/closure schedule	Normalized location/hours; manual entry works if location permission denied
4	Breakfast, lunch, evening snacks, dinner or custom slots; serving time/cutoff	Configured meal slots with overlap validation and local timezone
5	Choose Fixed-period, Meal-tracker, or Both; enable daily sales per slot if offered	Draft plan models and daily-meal capability
6	Durations, prices, included slots, credits, policies and optional discounts	Plan drafts; no publish before validation
7	Menu catalogue, base portions, special dishes, extras, substitutions and stock	Daily menu/plan inclusion draft with price snapshots
8	Offers, customer-facing preview, terms and commission acceptance	Offer drafts and recorded accepted agreement version
9	Submit business for review	Application status; admin review and audited decision

Review states

Draft → Submitted → In review → Approved / Needs changes / Rejected / Suspended. Only an approved and published business may appear in public discovery. Owners cannot self-approve or self-publish by editing a row.

Staff permissions

Role	May do	Should not do by default
Owner	Business, plans, menu, offers, team, reports and verified payout workflow	Bypass review or alter completed ledger entries
Manager	Daily menu, slot operations, attendance and customer issues	Owner transfer, bank change or unrestricted finance controls
Cashier	Display today's QR and see minimal redemption result	Plan pricing, payout, full customer profile
Menu editor	Items, portions, daily menu and sold-out status	Payments, settlements or role changes
Finance viewer	Scoped ledger and settlement report	Change prices or approve own sensitive payout

3. Plans, slots, menus and portions

Subscription models

Model	What customer buys	Redemption effect
Fixed-period	Access for 15 days, 1 month, 3 months, 6 months or 12 months	Record valid attendance; do not deduct a meal credit. Still enforce eligible slots and daily limits.
Meal-tracker	A defined number of meals plus an expiry date	Skip day = zero deduction. Successful eligible redemption = exactly one credit deducted. Stop at the first applicable limit.
Daily meal	A single meal in a configured date/slot	Create entitlement only after verified payment. Bind to customer, provider, date and slot; redeem once.

Pricing policy

Providers configure their own prices. A 15-day / 1 / 3 / 6 / 12-month price is a provider-approved value, not a system default. Longer-term discount savings must compare against a real approved reference price.

Meal slot configuration

- Name and code: breakfast, lunch, evening snacks, dinner or custom.
- Serving start/end, last redemption time, allowed weekdays, holidays and daily limits.
- Provider timezone and service date: use Asia/Kolkata for Indian operations.
- Prevent ambiguous overlapping slots; if overlap remains, return a clear slot choice rather than silently guessing.
- A mess with only lunch and dinner must not display breakfast to its customers.

Menu and extra rules

Rule type	Example	Customer display
Included base portion	Dal 1 serving, rice 1 serving, roti 4 pieces	Included in this plan
Plan-specific item	Paneer included only in Premium	Included with Premium only
Paid special dish	Paneer special +₹30	Optional extra +₹30
Extra quantity	Extra roti ₹5 each; rice ₹10 per serving	Quantity and unit price shown
Substitution / upgrade	Replace regular sabzi with a special dish for a set difference	Only available if provider enables it
Stock / sold out	Paneer stock reaches zero	Unavailable; no charge allowed

A base lunch/dinner redemption is one meal. Additional roti or a special dish is a separate line item and must not consume another meal credit. Save immutable item, quantity, plan-inclusion and price snapshots on completed records so future menu edits do not rewrite history.

4. Daily QR and provider dashboard

Daily QR workflow

- Backend issues one provider-specific QR session for the local service date. The same QR can stay visible at the counter through the day.
- QR payload must be signed/tokenized and bound to provider + service date; it cannot be a static business ID.
- Customer scans provider QR using the customer app. Provider app displays the QR; it does not scan a customer-generated QR in this product flow.
- Customer backend request validates QR, provider status, plan/daily entitlement, service date, current meal slot, limits, payment and duplicate-key rules.
- Menu preview shows plan-included items and paid extras. Sold-out extras cannot be charged.
- One atomic backend operation commits redemption and tracker credit deduction. UI success appears only after commit.
- Provider counter interface shows a privacy-minimal result: confirmed / already redeemed / invalid / expired / wrong slot / no entitlement / no credits.

Concurrency safeguard

Use a database unique key and transactional balance update for customer + provider + meal slot + service date. Concurrent scans cannot deduct twice. Same day QR may be used by many eligible customers, but each customer's allowed slot is redeemed only once.

Provider overview dashboard

Section	What to show	Avoid
Today at a glance	Configured slots, current slot status, verified redemptions when live	Invented meal counts or fake "busy" badges
QR card	Today's server-issued QR status and service date	Client-generated production QR or token secret exposure
Menu status	Today's published menu and sold-out items	AI-generated photos or unconfigured menu items
Plans	Active plans, renewals due and publication warnings	Hardcoded universal prices
Finance	Gross, platform/gateway fee, refund, payable and paid settlement separately	Gross sales presented as profit or settlement
Attention queue	Provider review, support and reconciliation exceptions	False green health status or fake ticket count

5. Admin and operations console

The admin panel is a restricted web console for the MESS team. The UI prototype includes overview, provider applications, providers, customers, plans/menus, redemptions, payments/settlements, offers, support, team/roles, audit log and system health. Every action must be connected to a trusted server endpoint before real operations.

Area	Key capabilities	Required control
Provider applications	Inspect owner, address, applicable licence/documents, menu/plan preview, agreement; approve/request changes/reject/suspend	Role check, reason, confirmation and audit entry
Customers	Minimum account/status details for support	Data minimization; scoped PII access
Plans & menus	Publication state, duration/price validation, portion rules, sold-out and policy errors	Server validates changes; preserve purchase snapshots
Redemptions	Search by provider/date/slot/customer; investigate duplicate and disputed scans	Read server-confirmed events; no silent ledger edits
Payments & settlements	Gateway status, webhook reconciliation, fees, refund, provider payable and payout status	Finance role; idempotency; maker/checker on high-risk actions
Offers	Effective dates, eligible plans, minimum order, limits and stacking	Server-side checkout validation
Support	Tickets, assignments, priority and resolution note	Scope access to the ticket and necessary user data
Team & roles	Invite/revoke staff, grant least privilege, MFA requirements	Server-side authorization and audit
Audit & system health	Append-only events, masked logs, migration/service status and feature flags	No secrets in browser; developer role read-only

Team roles

- Super Admin: role administration and high-risk platform configuration with MFA.
- Operations: provider verification, menu/publication issue and redemption/support investigation.
- Finance: reconciliation, refund and settlement workflow; use maker/checker where appropriate.
- Support: ticket management and the minimum necessary customer/provider data.
- Developer read-only: masked service logs, migrations and health status; no routine access to customer PII or payment credentials.

Authorization rule

Hidden buttons are not security. Every admin/provider operation must check identity, role, business ownership and allowed transition on the server.

6. Technical architecture and schema

Layer	Recommended implementation	Boundary
Provider mobile	Flutter, feature-based modules, shared theme and typed repository interfaces	UI does not decide plan eligibility, prices, approval or QR validity
Admin web	Next.js + TypeScript, responsive layout, accessible controls and protected server routes	Client role display is not authorization
Auth/data	Supabase Auth + PostgreSQL + private Storage, or reuse the existing backend after audit	RLS on every exposed table; test grants and policies
Domain services	Trusted functions/server endpoints for plan publish, QR issuance, redemption, payment and settlement	Atomic transactions, idempotency and server-authoritative state
Payments	Gateway adapter and sandbox first; signed webhooks and reconciliation	No credentials in Flutter/browser bundles

Core database groups

Group	Main tables
Identity / business	profiles, businesses, business_members, provider_documents, business_hours
Service / menus	meal_slots, menu_items, daily_menus, daily_menu_items
Pricing / subscriptions	plan_templates, plan_meal_slots, plan_included_items, subscriptions, daily_meal_orders
QR / attendance	daily_qr_sessions, meal_redemptions, redemption_addons
Finance / offers	payments, promotion_codes, promotion_redemptions, financial_ledger, provider_settlements
Customer / operations	business_feedback, support_tickets, platform_admin_roles, admin_audit_logs, platform_feature_flags

Key integrity requirements

- Money uses numeric/integer minor units, not floating point. Use immutable snapshots for price, commission terms, portions and discounts.
- Date-bound QR session is unique per provider/service date. Store token digest/metadata; return signed payload only through authorized backend flow.
- Unique/idempotency constraints protect gateway events and redemption events.
- A plan starts only after verified payment. Tracker credits, payment status, provider commission and payout are server-calculated.
- Apply explicit grants/revokes along with RLS. A policy does not automatically remove table-level grants.
- Private provider documents use a private storage bucket and short-lived signed access. Avoid storing raw secrets or unnecessary PII in logs.

7. Implementation order and release gates

Phase	Build	Exit criterion
0. Audit	Inspect existing repo, customer app, web assets, Supabase schema, env and working flows	Integration checklist; no destructive changes
1. Domain model	Provider, slots, menu portions, plans, entitlements, extras, offers	Server rules and typed models reviewed
2. Auth/onboarding	Provider login/register, business/location draft, document upload and review status	Owner and business access tests pass
3. Menu/plans	Slots, weekly/daily menus, plan models/durations/price and included/extra items	Customer preview matches server validation
4. Admin console	Secure auth/roles, provider review, support, audit and health views	Role isolation and audit tests pass
5. QR/redemption	Provider daily QR, customer scan, slot resolution and atomic transaction	Duplicate/concurrency tests pass
6. Payments	Daily meal and add-on/subscription checkout, signed webhooks, refunds and ledger	Sandbox reconciliation + idempotency pass
7. Reports/support	Attendance, earnings, settlement, feedback and support workflows	Live queries, access restrictions and empty/error states verified
8. Release	Accessibility, responsive checks, docs, backup/rollback and pilot	All critical tests run; open blockers disclosed

Must-pass acceptance checks

- Provider cannot self-approve or publish a production plan/menu by changing a client field.
- Fixed-period plan records valid meal attendance but does not decrease meal credits.
- Tracker plan deducts exactly one credit after successful redemption; skipped days deduct zero.
- Daily meal cannot redeem without verified payment and its customer/provider/date/slot entitlement.
- Same provider QR works for eligible customers that day; a customer cannot redeem the same allowed slot twice.
- Wrong provider/date/signature, expired QR, out-of-slot, no credits, sold-out extra and failed payment have clear errors and no unintended state change.
- Admin finance/role actions are server-authorized, reasoned and audited. Developer views are read-only and masked.
- No dead actions, broken links, page-level horizontal overflow, fake metrics/reviews/photos or exposed secrets.

Prototype status

The ZIP contains source prototypes, not a deployed platform. Flutter SDK was not available in the authoring environment; run flutter analyze/test/build locally. Install web dependencies and run Next.js typecheck/lint/build. Treat the SQL as a reviewable baseline and validate it in a staging database before any live migration.

Next handoff: extract MESS_Provider_Admin_Package.zip, open the existing MESS repository in Antigravity, paste prompts/ANTIGRAVITY_MASTER_PROMPT.md, and require iterative implementation with actual test results. Use LOOP_CONTINUE_PROMPT.md after each phase and RELEASE_AUDIT_PROMPT.md before launch.